

Indivisible Goods Recap

Definition 1. We have a set $N = [n]$ of players and a set $\mathcal{M}$ of m goods. Each player i has a nonnegative, additive valuation $v_i(S) = \sum_{g \in S} v_i(g)$. An allocation $A = (A_1, \dots, A_n)$ is a partition of $\mathcal{M}$ into disjoint bundles, one per player. Empty bundles are allowed; all goods are allocated.

For unnormalized values, proportionality then asks for $v_i(A_i) \geq v_i(\mathcal{M})/n$. Envy-freeness cannot always be achieved, so we have relaxed notions.

Definition 2. For an allocation $A = (A_1, \dots, A_n)$:

1. The allocation is *envy-free up to one good* (EF1) if for every i, j with $A_j \neq \emptyset$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for some } g \in A_j.$$

2. The allocation is *envy-free up to any good* (EFX) if for every i, j ,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for every } g \in A_j.$$

An empty bundle is never envied. Removing a good is only a comparison: the actual allocation still contains all the goods. Here EFX uses every good, including zero-valued goods.

EFX $\implies$ EF1.

Two-Player Cut and Choose

Exercise. When can we achieve an EF1 allocation? Try $n = 2$ players, any m goods, and additive valuations. Use the following values to illustrate your rule.

Player	A	B	C
1	10	9	2
2	1	8	4

Round Robin

The Round Robin Mechanism. First, set an arbitrary order of the players, denoted $1, 2, \dots, n$. Until all the goods are allocated, go through the players in this order repeatedly. On player i 's turn, she chooses her most valuable remaining good. Ties may be broken arbitrarily.

Theorem 1. *For nonnegative additive valuations, round robin produces an EF1 allocation.*

Run round robin on the table above, with player 1 choosing first. Check the EF1 condition in both directions.

How Unfair Can Round Robin Be?

Exercise. There are $n \geq 2$ players with identical additive values: $n - 1$ goods worth n each and n goods worth 1 each. Run round robin in the order $1, \dots, n$. Compare with another allocation. Is the round-robin outcome EFX?

Maximin Share

Suppose a player could partition the goods into n bundles, but had to take her least valuable bundle. What value could she guarantee herself?

Definition 3. The *maximin share* of player i is

$$\text{MMS}_i = \max_{(B_1, \dots, B_n) \in \Pi_n(\mathcal{M})} \min_{1 \leq j \leq n} v_i(B_j),$$

where $\Pi_n(\mathcal{M})$ is the set of partitions into n bundles, possibly empty. An allocation A is *MMS-fair* if $v_i(A_i) \geq \text{MMS}_i$ for every player i .

Whose valuation is used for all n bundles in this definition? How does MMS_i compare with the proportional share $v_i(\mathcal{M})/n$?

Exercise. In the example with $n - 1$ goods worth n and n goods worth 1, compute each player's MMS. What fraction of her MMS does the last player receive under round robin?

Rent Division

Housemates disagree about which room is best. Prices let us compensate the housemate assigned a less desirable room. An envy-free outcome makes each person prefer their own room at its price to every alternative.

There are n housemates, n rooms, and total rent R . Housemate i values room j at $v_{ij} \geq 0$, with $\sum_j v_{ij} = R$ for each i . Utilities are quasilinear; prices may be negative unless we explicitly require otherwise.

Definition 4. An *outcome* is a pair (σ, p) , where the assignment σ is a bijection from housemates to rooms and the price vector $p = (p_1, \dots, p_n)$ satisfies $\sum_j p_j = R$ (the landlord gets paid, exactly). The utility of housemate i is $u_i = v_{i\sigma(i)} - p_{\sigma(i)}$.

Definition 5. An outcome (σ, p) is *envy-free* if for every housemate i and every room j ,

$$v_{i\sigma(i)} - p_{\sigma(i)} \geq v_{ij} - p_j. \tag{1}$$

That is, nobody wants to trade places with anybody else, where trading places means swapping both the room *and* the rent that comes with it.

Two Rooms

Two housemates have total rent $R = 100$ and the following values:

Housemate	Room A	Room B
1	70	30
2	60	40

Assign room A to housemate 1 and room B to housemate 2. Let $p_A = q$, so $p_B = 100 - q$. For which prices is this assignment envy-free?

Write each housemate's no-envy inequality and find all feasible values of q .

Do equal rents work? What changes when we can attach prices to indivisible rooms?

Fairness and Efficiency

Envy-freeness compares each person's room with all the alternatives at the same prices. Does it also tell us which assignment maximizes total value?

Theorem 2. *Let (σ, p) be an envy-free rent division outcome. Then σ maximizes total value: for every other assignment τ ,*

$$\sum_i v_{i\sigma(i)} \geq \sum_i v_{i\tau(i)}.$$

Proof. Fix another assignment τ , keeping each room's price fixed. Compare housemate i 's assigned room with $\tau(i)$, then sum over housemates.

Does a welfare-maximizing assignment together with any prices summing to R have to be envy-free?

Find a counterexample.

Choosing within the Interval

Return to $R = 100$ and values $(70, 30)$ and $(60, 40)$. Housemate 1 gets A , housemate 2 gets B , and $p_A = q$. Fill in the prices and utilities.

q	p_A	p_B	u_1	u_2
60				
65				
70				

What stays fixed as prices change? Does envy-freeness select one outcome?

Now take general two-person values with $v_{i1} + v_{i2} = R$, and assign housemate i to room i . Derive the interval of envy-free prices p_1 , using $p_2 = R - p_1$.

If the interval is nonempty, what utilities result from choosing its midpoint?